

# Luke Christian Barter

Corpus Christi, Texas, United States | [lukechris.barter93@gmail.com](mailto:lukechris.barter93@gmail.com) | +1 (915) 293-4676

## Summary:

---

- I am a Data Scientist / ML Engineer with over 10 years of professional experience specializing in Data Engineering, Natural Language Processing (NLP), Machine Learning (ML), Deep Learning (DL), and Computer Vision.
- My expertise spans LLM integration, Agentic AI systems, and multi-agent orchestration using tools like LangChain, AutoGen, CrewAI, and LangGraph, enabling autonomous reasoning, retrieval, and task execution.
- Skilled in RAG architectures, semantic/knowledge graph search, and Edge AI deployment using quantized LLMs for efficient real-time inference.
- Strong focus on data security and compliance, implementing HIPAA-aligned ML pipelines, secure model APIs, and role-based data access controls across healthcare and enterprise systems.
- Experienced in data engineering and cloud analytics with Azure, GCP, and AWS, building scalable ETL workflows, MLOps pipelines, and predictive models across large datasets.
- Proficient in Python and SQL, with deep expertise in TensorFlow, PyTorch, Scikit-learn, Pandas, and modern visualization tools including Power BI, Tableau, and Streamlit to deliver clear, actionable insights.
- Passionate about building intelligent, production-grade AI solutions that combine autonomy, compliance, and performance to drive measurable business impact.

## Skills:

---

- **Specialized in:** Machine Learning, Deep Learning, Statistical Modeling, Data Analysis, Predictive Analysis, Data Extraction, Data Manipulation, Data Transformation, Data Mining, Data Visualization and Business Intelligence and LLM.
- **Deep Learning:** TensorFlow/Keras, PyTorch, FFNN, NTK, GANs, Transfer learning, Auto encoders, CNN, RCNN, Fast & Faster RCNN, Masked RCNN, YOLO, Transformers, Word Embeddings, Bidirectional RNN, LSTM, Attention Models, BERT, GPT
- **Machine learning:** Supervised, Unsupervised, Anomaly Detection, Time series algorithms including Regression, Binary/Multiclass Classification, Bagging, Boosting, Regularization, Moving average, Ensembles, Dimensionality Reduction
- **Computer Vision:** OpenCV, Tesseract, OCR
- **Python libraries:** pandas, NumPy, BeautifulSoup, matplotlib, SciPy, bokeh, seaborn, scikit-learn, PyOD, XGBoost, Jupyter Notebook/Colab, Spark ML, GIT, DASK, CUDA
- **Agentic AI & Multi-Agent Systems:** AutoGen, CrewAI, LangGraph, LangChain Agents, N8N, tool-use orchestration, memory-driven agents.
- **RAG & Semantic Search:** Pinecone, FAISS, Weaviate, Chroma, Neo4j, Knowledge Graphs, ElasticSearch.
- **Edge / Lightweight AI:** Quantized LLMs, MLC, Ollama, GGUF, model compression, on-device inference.
- **Visualization & BI:** Power BI, Tableau, Streamlit, Plotly, Dash.m
- **Cloud Platforms:**
  - Microsoft Azure:** Azure ML, ADF, Databricks, Azure Functions, ADLS, SQL DB, Synapse, Cosmos DB, Service Bus, EventHub, CLI, Terraform, DevOps CI/CD, PowerBI, VSCode.
  - Google Cloud Platform:** Big table, Big Query, Dataflow, Airflow
  - Amazon Web Services :** EC2, RDS, EBS, S3, Lambda, Cognito, Rekognition, SageMaker
- **Cloud Security:** IAM, VPC, Key Vault, KMS, Secret Manager, TLS/SSL, encryption-at-rest, encryption-in-transit, RBAC, HIPAA/GDPR compliance
- **Big Data:** SPARK, Scala, Java, Hadoop, Hive, Impala, Oozie, Zeppelin, Hue, Kafka, MapReduce
- **Languages:** Python, PySpark, R, Java, C++, JavaScript(React.js, Node.js), Scala, C#, SQL
- **Open-Sources:** GPT, Hugging Face,
- **Database:** Greenplum DB, SQL, Postgres SQL, MongoDB

## Education:

---

Texas A&M University, Bachelor of Mathematics and Computer Science, Mar 2010 - May 2014

## Experience:

---

### Gen AI / NLP Engineer (Remote)

#### Tenet Healthcare: Mar 2020 - Aug 2025 (5 yrs 6 mos)

- Created interactive web-based user interfaces for AI models using tools like Gradio and Streamlit.
- Designed and implemented RAG architecture leveraging Pinecone for real-time vector similarity searches to enhance the accuracy of generated responses.
- Developed Telegram bots that can assist and guide members to learn about the specific projects using LLMs.
- Designed and maintained scalable ETL pipelines using PySpark and Azure Databricks, implementing data encryption, access policies, and PHI masking for over 10TB of structured and semi-structured healthcare data.
- Led R&D for LLM-based content generation tools used in a media automation platform; leveraged Hugging Face Transformers and OpenAI APIs to build and fine-tune models for creative writing, video scripting, and summarization.
- Designed and deployed agentic AI workflows using ChatGPT and N8N to automate sales and customer engagement processes, reducing manual touchpoints by 60%.
- Used LeanIX to map business-process-to-application dependencies, supporting a smooth ERP cloud migration with minimal disruption.
- Integrated GenAI models with Salesforce and Gong, enabling dynamic call summarization, sentiment tagging, and personalized follow-up recommendations.
- Designed and deployed customer segmentation models using KMeans and DBSCAN to identify underserved segments, improving targeted marketing ROI by 28%.
- Optimized large-scale SQL queries in BigQuery to analyze 500M+ rows of transactional data, reducing query time by 65% using partitioning and clustering techniques.
- Developed AI agents orchestrated with AutoGen, CrewAI, and LangGraph to automate report generation, data retrieval, and compliance workflows.
- Integrated Knowledge Graph / Semantic Search pipelines (Neo4j + OpenAI embeddings) for contextual query resolution and medical terminology mapping.
- Implemented Edge / Lightweight AI inference using quantized LLMs via MLC and Ollama, enabling low-latency healthcare deployments.
- Developed metadata-driven Python automation pipelines for SDTM dataset generation, reducing manual programming time by 70% across multiple studies.
- Integrated LangChain with various data sources and APIs to provide real-time, context-aware responses.
- Leveraged RAG to combine retrieval-based and generation-based approaches, improving the quality of generated responses in a chatbot application.
- Collaborated with InfoSec teams to establish model governance, encryption-at-rest, and network isolation in Azure ML and AWS SageMaker environments.
- Developed a scalable personalization engine using embedding models + vector search (e.g., FAISS / Pinecone), deployed on AWS SageMaker and integrated into a live content delivery pipeline.
- Supported cloud transformation strategy by leveraging LeanIX's cloud readiness assessments and generating visual heatmaps for exec buy-in.
- Trained CNN-LSTM architectures for image-text multimodal document processing (e.g., checks, IDs), achieving 95% IoU for bounding box prediction and 90% OCR accuracy.
- Developed optimized data models and star schemas in Azure Synapse Analytics (dedicated + serverless pools) for enterprise-wide reporting.
- Collaborated with a team of data scientists and NLP experts to define the fine-tuning objectives, data requirements, and evaluation metrics for successful model adaptation.
- Built NLP models for controlled terminology mapping (MedDRA, CDISC) using spaCy and scikit-learn, achieving over 90% accuracy.
- Led end-to-end ML product lifecycle from prototype to production using MLflow and Dockerized pipelines on AWS SageMaker and ECS.
- Built a multi-agent coordination framework (A2A) to handle lead routing, data enrichment, and follow-up across Outreach, Salesforce, and internal CRMs.
- Built end-to-end ML pipelines on Databricks and GCP, using PySpark for preprocessing and MLflow for model tracking and versioning.
- Developed and maintained Python-based frameworks for automated Define.xml generation and SDTM conversion from raw data.
- Developed interactive Streamlit dashboards to visualize model predictions, feature importances, and SHAP explanations- improving stakeholder buy-in and model transparency.
- Developed APIs to integrate Pinecone-powered RAG systems into client applications.
- Designed and implemented a robust face recognition system using OpenCV and Dlib for real-time authentication.

- Built ML pipelines using BigQuery ML to train logistic regression and time series forecasting models directly within BigQuery without external tools.
- Collaborated cross-functionally with product managers and domain experts to convert complex insights into intuitive dashboards using Plotly and Streamlit.
- Developed and maintained scalable server-side applications using Node.js, Express, and MongoDB, leading to a 30% increase in application performance and stability.
- Delivered clear, actionable ML insights to non-technical stakeholders through visual storytelling and iterative feedback cycles.
- Created reusable Python libraries to standardize domain identification and data transformation logic aligned with CDISC/SDTM rules.
- Automated CI/CD workflows using Azure DevOps, integrating unit tests and data validation steps for production-grade pipelines.
- Applied fine-tuning techniques to pre-trained language models such as GPT-3, BERT, or RoBERTa to adapt them for specific natural language processing (NLP) tasks.
- Developed and integrated Gen AI-based chatbots into the company's customer service platform, enhancing user experience with natural, conversational responses.

#### **Data Engineer / Data Scientist (Remote)**

**Arena: Aug 2015 - Jan 2020 (4 yrs 9 mos)**

- Leveraged PySpark to perform large-scale data mining and statistical analysis on customer behavior data.
- Built LSTM and Prophet models for transactional time-series forecasting, improving financial planning accuracy by 25% for client accounts.
- Led design and deployment of AI/ML pipelines to automate clinical data standardization workflows across ADaM and SDTM datasets.
- Built automated GenAI experimentation workflows using GCP Vertex AI and Azure ML, including performance evaluation on real-world data and A/B testing to validate model-driven creative optimization.
- Integrated LeanIX with ServiceNow CMDB to automate real-time updates of application metadata and ownership assignments.
- Designed and maintained large-scale ETL pipelines using PySpark, Airflow, and GCP Dataflow, processing 10 TB of transactional and clinical data daily.
- Built Power BI and Tableau dashboards connected to BigQuery and Azure Synapse, visualizing model predictions, KPIs, and trends for executive teams.
- Applied Prompt Engineering and Function Calling with OpenAI APIs to build internal NLP assistants for automated data query and labeling workflows.
- Defined TensorFlow-based text classification models for PHI detection, integrating data redaction and PII anonymization into the model workflow, reducing compliance risk by 90%.
- Utilized Python libraries such as Pandas, NumPy, and Scikit-learn for data manipulation, analysis, and model development.
- Applied machine learning techniques to classify raw clinical terms and automate dataset domain assignments, reducing QC time by 60%.
- Built a Streamlit-based self-service analytics tool for business users to slice and analyze marketing campaign data, cutting ad-hoc request volume by 40%.
- Developed command-line tools and scripts using Node.js, automating repetitive tasks and improving developer workflow.
- Deployed real-time and batch inference solutions using SageMaker, improving the operational efficiency of predictive analytics.
- Implemented data storage solutions with MongoDB and PostgreSQL, ensuring efficient data handling and retrieval.
- Integrated Azure AI Search with client applications through REST APIs and SDKs.
- Prototyped and Dockerized an attention-based model for document classification and deployed using the Kubernetes command line.
- Supported implementation of AI-driven decision support tools for mapping case report form data to SDTM variables and domains.
- Developed a mobile OCR system utilizing YOLOv7, CRNN, and Mobile BERT models, achieving 95% accuracy in extracting information from diverse ID card and driving licenses.
- Deployed real-time and batch inference solutions using SageMaker, improving the operational efficiency of predictive analytics.
- Implemented I2 statistical and machine learning solutions end-to-end through the life cycle of deploying and maintaining metadata at scale.
- Developed a sentiment analysis model to help the marketing team understand customer feedback, resulting in a 10% increase in customer satisfaction.
- Conducted code reviews and pair programming sessions to ensure high-quality Terraform code.
- Integrated Terraform with CI/CD pipelines using tools like Jenkins, GitLab CI, and GitHub Actions to automate infrastructure deployment.
- Designed and deployed APIs via Flask to serve predictions in a cloud-native microservices architecture, using Docker, CI/CD (GitHub Actions), and Terraform for infrastructure automation.